

K.D.S.H.M. Perera

B.Sc. Engineering (Honours)

99B, Colombo Road, Pokunuwita, Sri Lanka

📞 +94 721 982 710 | 🎓 | 📺 | 🔄 | 🌐 | ✉️ heshanm@uom.lk

Profile

My primary research interest lies in bio-inspired robotics, with a particular focus on unconventional locomotive platforms capable of navigating complex, unstructured environments. My work spans the full development pipeline of a robotic system, including mechanical design, hardware prototyping, embedded systems, mathematical modeling, simulation, and dynamic control. I am further interested in scaling these principles through swarm frameworks, deploying decentralized collectives of bio-inspired robots to solve complex tasks in unpredictable real-world settings.

Education

- 2021 – 2025 **B.Sc. Engineering (Hons) in Mechanical Engineering**
Specialisation: Mechatronic Systems Engineering, *University of Moratuwa*, Sri Lanka
GPA: 3.79 / 4.00
Class Rank: 10/120
Minor: Mathematics
- 2017 – 2019 **GCE Advanced Level, Physical Science Stream**, *D.S. Senanayake College, Colombo 07*
Results: 2 A's & 1 B; Z-Score: 2.3409

Work Experience

- Feb 2025 – Present **Temporary Lecturer**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka
- Aug 2025 – Jan 2026 **Temporary Instructor**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka
- Mar 2024 – May 2024 **Intern / Student Researcher**, *Shandong University of Science and Technology*, Shandong, China
- Nov 2023 – Feb 2024 **Intern / Project Engineer**, *Dipped Product PLC*, Sri Lanka

Research Projects

- Underwater Drone for Coral Observation 📺 Contributed to the design and development of a shallow-sea underwater drone using a novel fuzzy-adaptive sliding-mode controller for robust trajectory tracking; developed a VR-based teleoperation interface for real-time control and visualization.
- Brachiating Robot 📺 Designed, simulated, and built a prototype brachiation robot to investigate dynamic locomotion and control across variable bar spacing and heights, using a simplified control strategy where only the reaching hand is actively guided toward the next target, achieving brachiation in a single swing.
- Biomimetic Fish Robot (Ongoing) Developing a soft tendon-driven biomimetic fish robot featuring a non-contact magnetic actuation interface.

Academic Projects

Environment Monitoring Station 	Implemented real-time environmental monitoring with signal conditioning, processing, remote communication and data visualization (web/mobile).
App-Controlled Mobile Robot	Developed a differential-drive mobile robot with image-based object detection, path planning, and manual/autonomous operation via a mobile app.
Micro-ROS Based Home Automation	Led development of an IoT smart-home prototype integrating micro-ROS for autonomous temperature-based fan control and web-based lighting/fan management.
Smart Garbage Collector	Collaboratively engineered a mobile robot using image processing to autonomously navigate predefined routes and collect litter.
Modular Microconveyor	Designed a modular microconveyor using chevron electrothermal actuators and a serpentine beam ratchet for precise, incremental motion; validated with multiphysics simulation.

Industrial Projects

Dynamic Work Bar Transfer Mechanism	Designed a pneumatic system to halt and transfer work bars between leaching tanks.
Web-Based SCADA Visualization	Developed a full-stack web application for real-time SCADA dashboards and historical data visualization.
Hydraulic Lifting Platform	Researched and prototyped a hydraulic lift for wheelchair accessibility (Shandong University of Science and Technology).

Achievements

Best Paper Award	Robotics & Intelligent Systems Track — Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka.
The Institution Best Project	Institution of Mechanical Engineers (IMechE) — University of Moratuwa, Sri Lanka (2025).
Dean's List	Semesters 1, 2, 5, 7

Research Publications

- (First author & presenter) *Brachiation Robot Inspired by Gibbon Dynamics: Insight for Control Across Irregular Bar Spacing and Heights*. Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka. **Best Paper Award, Robotics & Intelligent Systems**. DOI: 10.1109/MERCon67903.2025.11217001.
- (First author & presenter) *A Fuzzy-Adaptive Sliding-Mode Control Strategy for Optimizing Control of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka. DOI: 10.1109/MERCon67903.2025.11217001.
- (Co-first author) *Development of Marine Grade Acrylonitrile – Butadiene – Styrene (ABS) based Hull of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka. DOI: 10.1109/MERCon67903.2025.11217039.
- (Co-first author) *CFD-Driven Hydrodynamic Optimisation of a Whale Shark-Inspired Underwater Drone for Shallow Reef Observation*. 12th International Conference on Heat Transfer and Fluid Flow (HTFF 2025). DOI:10.11159/htff25.243.

- (First author) *A Chevron-Actuated Microconveyor for Precise Wafer-Scale Microfactory Transportation*. Mechanical Engineering Research Symposium 2025 (MERS 2025), Department of Mechanical Engineering, University of Moratuwa. Accepted.
- (Co-first author) *WS Ray: Extending Human Perception with a Teleoperated Underwater Drone for Subaquatic Environments*. International Conference on Automation Science and Engineering (CASE 2026). Under review.

Skills

Programming	Python, C++, MATLAB & Simulink , LabVIEW
Robotics & Simulation	ROS2, micro-ROS, Gazebo, MuJoCo, Webots, OpenCV
Design & Analysis	SolidWorks, AutoCAD, Ansys, Adams, Fluidsim
Embedded Systems & Electronics	Atmel Studio, Proteus, Arduino & ESP32, Raspberry Pi
Web & Data	HTML/CSS, JavaScript, SQL, Flutter
Office & Graphics	LaTeX, Microsoft Word, PowerPoint, Excel, Adobe Illustrator, Photoshop

Languages

Sinhala	Native
English	Fluent (TOEFL iBT: 5.0/6.0)

Mentorship & Service

2026 Committee Member	Organizing Committee, MERS 2026, University of Moratuwa
2026 Lecturer	ME4740: Advanced Topics in Mechatronic System Engineering – Modeling & Simulation Section
2026 Lecturer	ME4191: Robotics & Automation Systems – ROS 2 Workshop
2026 Team Mentor	Singapore Autonomous Underwater Vehicle Challenge (SAUVC 2026)
2025 Instructor	ME3261: Mechatronic System Design Project – Original ROS 2 Workshop
2025 Instructor	Mechanical & Mechatronic Engineering Courses, University of Moratuwa
2023 Committee Member	Logistics Committee, EXMO 2023, University of Moratuwa
2023 Committee Member	Organizing Committee, MechatronX Inter-School Robotics Competition
2019 Member	Danuma Yathra Community Service Project
School Level Member	Science Quiz Team Organizer, D.S. Senanayake College

References

Prof. Ranjith Amarasinghe

Head of Department
Department of Mechanical Engineering
University of Moratuwa
Tel: +94 71 8517882
Email: ranama@uom.lk

Dr. Dumith Jayathilaka

Stream Coordinator
Mechatronic Systems Engineering
University of Moratuwa
Tel: +94 11 2650301 ext. 4522
Email: dumithj@uom.lk